Installation Guide

SAP Landscape Transformation Replication Server

Document Version: 2.1 – 2017-08-23

CUSTOMER

Operational Data Provisioning in Real-time with SAP Landscape Transformation Replication Server

Typographic Conventions

Type Style	Description
Example	Words or characters quoted from the screen. These include field names, screen titles, pushbuttons labels, menu names, menu paths, and menu options. Textual cross-references to other documents.
Example	Emphasized words or expressions.
`EXAMPLE`	Technical names of system objects. These include report names, program names, transaction codes, table names, and key concepts of a programming language when they are surrounded by body text, for example, SELECT and INCLUDE.
`Example`	Output on the screen. This includes file and directory names and their paths, messages, names of variables and parameters, source text, and names of installation, upgrade and database tools.
`Example`	Exact user entry. These are words or characters that you enter in the system exactly as they appear in the documentation.
`<Example>`	Variable user entry. Angle brackets indicate that you replace these words and characters with appropriate entries to make entries in the system.
`EXAMPLE`	Keys on the keyboard, for example, `F2` or `ENTER`.

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Document History

Version	Date	Change
1.0	2013-09-12	Initial Version
1.1	2013-09-24	Updates to reflect SAP Landscape Transformation Replication Server DMIS_2011 support package 05
1.2	2014-01-29	Updates to reflect SAP Landscape Transformation Replication Server DMIS_2011 support package 06
1.3	2014-07-08	Updates to reflect SAP Landscape Transformation Replication Server DMIS_2011 support package 07
1.4	2014-07-30	Enhancements to section 5.2, User Creation and Connections and section 5.6, Data Replication Triggered from Subscriber System.
1.5	2015-01-21	Updates to reflect SAP Landscape Transformation Replication Server DMIS_2011 support package 08
1.6	2015-07-28	Updates to reflect SAP Landscape Transformation Replication Server DMIS_2011 support package 09
1.7	2015-08-10	Updated with additional information about SAP NetWeaver Business Client
1.8	2015-12-28	Updates to reflect SAP Landscape Transformation Replication Server DMIS_2011 support package 10
1.9	2016-06-10	Updates to reflect SAP Landscape Transformation Replication Server DMIS_2011 support package 11
2.0	2017-01-16	Updates to reflect SAP Landscape Transformation Replication Server DMIS_2011 support package 12
2.1	2017-06-14	Updates to reflect SAP Landscape Transformation Replication Server DMIS_2011 support package 13
2.2	2017-07-23	Update to include details of SAP Note 2520600, which is mandatory for DMIS 2011 SP13.

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1 Getting Started

1.1 About this Document

Purpose

This guide details the installation and configuration of the SAP Landscape Transformation Replication Server in order to facilitate real-time data replication for the Operational Data Provisioning framework.

This guide is intended for system administrators and consultants performing and initial install and configuration of SAP LT Replication Server. Proficiency with SAP NetWeaver Basis is required to complete the installation.

This guide will take you through the required steps to:

- Decide on a suitable installation type based on the existing system landscape
- Install the SAP LT Replication Server
- Configure the source data system for RFC access from the SAP LT Replication Server
- Configure the SAP LT Replication Server as your target system
- Connect a subscriber system to the operational Delta Queue in the SAP LT Replication Server system
- Start replication of DataSources from the subscriber system

1.2 Related Information

1.2.1 Planning Information

For more information about planning topics not covered in this guide, see the following content:

Content	Location
Latest versions of installation and upgrade guides	http://service.sap.com/instguides
Sizing, calculation of hardware requirements - such as CPU, disk and memory resource - with the Quick Sizer tool	http://service.sap.com/quicksizer

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Content	Location
Released platforms and technology-related topics such as maintenance strategies and language support	http://service.sap.com/platforms To access the Platform Availability Matrix directly, enter http://support.sap.com/pam
Network security	http://service.sap.com/securityguide
High Availability	http://scn.sap.com/docs/DOC-7848
Performance	http://service.sap.com/performance
Information about Support Package Stacks, latest software versions and patch level requirements	http://support.sap.com/sp-stacks
Information about Unicode technology	http://scn.sap.com/community/internationalization-and-unicode
Content	Location

1.2.2 Further Useful Links

The following table lists further useful links:

Content	Location
Information about creating error messages	http://support.sap.com/incident
SAP Notes search	http://support.sap.com/notes
SAP Software Distribution Center (software download and ordering of software)	http://support.sap.com/swdc
SAP Online Knowledge Products (OKPs) – role-specific Learning Maps	http://service.sap.com/rkt

1.2.3 Related Documentation and Guides for Operational Data Provisioning

For more information about the Operational Data Provisioning framework as part of SAP NetWeaver, see the resources listed in the table below.

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Topic	Guide/Tool	Quick Link
Operational Data Provisioning	SAP Help Portal (http://help.sap.com/) → Technology Platform → SAP NetWeaver →SAP NetWeaver Platform → SAP NetWeaver 7.4 → Application Help → Function-Oriented View → Search and Operational Analytics → Operational Data Provisioning	See here.
Transferring Data with SAP Landscape Transformation Replication Server	SAP Help Portal (http://help.sap.com/) → Technology Platform → SAP NetWeaver →SAP NetWeaver Platform → SAP NetWeaver 7.4 → Application Help → Function-Oriented View → SAP Business Warehouse → Data Warehousing → Modeling → Data Acquisition Layer → Data Provision Using Source Systems → Transferring Data with SAP Landscape Transformation Replication Server	See here.

1.2.4 Related SAP Landscape Transformation Replication Server Guides

For information about additional SAP LT Replication Server guides and resources, see SAP Note 1577441 (the central SAP Note for SAP LT replication server installation).

1.3 Key Terms

The following table contains key terms related to the SAP LT Replication Server:

Term	Definition
Configuration	The definition of the parameters that the SAP LT Replication Server uses to replicate data from one or more source systems to one or more target systems. The configuration specifies the source system, the target system, and the relevant connections.
Configuration and Monitoring Dashboard	An application that runs on the SAP LT Replication Server that you use to specify configuration information (such as the source and target systems, and relevant connections) so that data can be replicated. You can also use it to monitor the replication status.
Database trigger	A database trigger is procedural code that is automatically executed in response to certain events to a particular database table or view.

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Term	Definition
Data transfer Job	A job that is used for the data transfer process in the SAP LT Replication Server.
Initial load	A step within the trigger-based replication process that loads data from the source system to target system.
Initial load Job	A job that is used for the initial load process in the SAP LT Replication Server.
Logging table	A table in the source system in which any changes to a table that is being replicated are recorded. This ensures that the SAP LT Replication Server can replicate these changes to the target system.
Master job	A monitoring job that calls the initial load jobs and the data transfer jobs in the SAP LT Replication Server.
Reading type	A technique for reading data from tables in the target system during the initial load process.
SAP LT Replication Server	An SAP application that facilitates the replication of data from one or more source systems to one or more target systems. The source systems can be ABAP-based or non-ABAP-based systems.
Trigger-based replication	A technique for replicating data where an initial load is first performed that loads data from the source to the target system, and a replication phase begins whereby only changes to the source database (recorded by databases triggers) are replicated to the target database, thereby facilitating data replication in real-time.
DataSource	Object that makes data for a business unit available to SAP BW. A DataSource contains a number of logically-related fields that are arranged in a flat structure and contain data to be transferred into SAP BW.
Configuration	The definition of the parameters that the SAP LT Replication Server uses to replicate data from one or more source systems to one or more target systems. The configuration specifies the source system, the target system, and the relevant connections.
Configuration and Monitoring Dashboard	An application that runs on the SAP LT Replication Server that you use to specify configuration information (such as the source and target systems, and relevant connections) so that data can be replicated. You can also use it to monitor the replication status.
Subscriber	An SAP application (currently SAP BW or SAP Business Objects Data Services), that connects to the API of the Operational Data Provisioning Framework of the SAP LT Replication Server system in order to consume data.

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1.4 Important SAP Notes

You must read the following SAP Notes before you start the installation. These SAP Notes contain the most recent information on the installation, as well as corrections to the installation documentation.

Make sure that you have the up-to-date version of each SAP Note, which you can find on SAP Support Portal at http://support.sap.com/notes.

SAP Note Number	Title	Description
2520600	After upgrade to DMIS 2011 SP13: Tables automatically stop and vanish from ODQ-configurations	Mandatory SAP Note for SP13.
2120574	Operational Data Provisioning with SAP LT Replication Server	This SAP Note describes the ODP scenario with SAP LT Replication Server and provides installation details.
1931427	Operational Data Provisioning API 2.0	Collective Note with Corrections for ODP API 2.0
19466	Downloading SAP Kernel patches	Downloading a kernel patch in the Software Distribution Center.
517484	Inactive services in the Internet Communication Framework	The Internet Communication Framework Services are inactive when you install the SAP Web Application Server.
2446237	Installation/Upgrade SLT - DMIS 2011 SP13	This SAP Note describes the installation or upgrade of SAP LT Replication Server to the relevant DMIS SP.
1817467	Enable push of data units for Standard & DeltaInit requests	This is a corrective note for ODQ functionality which has to be implemented with DMIS2011 SP5 to use the ODP scenario.

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2 Planning

2.1 Introduction

Important Information

If you are running SAP Landscape Transformation Replication Server on DMIS 2011 SP13 (or you have just upgraded to DMIS 2011 SP13), you must implement SAP Note 2520600 before using any configuration to replicate data.

SAP Operational Data Provisioning (ODP) is a solution which is integrated with SAP NetWeaver (release 7.30 onwards). It is a framework that allows for storing data in a queue, the Operational Delta Queue (ODQ) and makes the data available to SAP applications for extraction and replication by means of the ODP API. Data can be delivered to the queue with processes or extractors; the queue is a staging area where the data is stored, and the SAP applications subscribe to the API and consume the data from the queue. The ODP is capable of replicating delta data, and now in combination with SAP LT Replication Server ODP is also capable of replicating data in real-time. It is also possible to replicate pool and cluster tables, as well as data and views from non-ABAP source systems.

Minimum SAP NetWeaver Requirements:

This ODP scenario is available with the following SAP NetWeaver releases:

- NW7.30 SP10 or with SP8 (together with SAP Note 1848320)
- NW7.31 SP8 or SP3 (together with SAP Note 1817467)
- NW7.40 SP4 or RTC (together with SAP Note 1717467)

Current Restrictions:

- No replacement for complex extractors with delta logic

The following components are used in the technical system landscape:

- SAP source system

  The source system tracks database changes by using database triggers. It records information about changes in the logging tables. Read modules (located on the SAP source system) transfer the data from the source system to the SAP LT Replication Server. The relevant data is read from the application tables.

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- SAP LT Replication Server System

  An SAP system that facilitates the replication of data from one or more source systems
- Target System

  For the ODP scenario, the SAP LT Replication Server system acts as the target system, storing all replicated data in the Operational Delta Queue-of the Operational Data Provisioning Framework.
- Subscriber

  A subscriber is an SAP application (currently SAP BW or SAP Business Objects Data Services), that connects to the API of the Operational Data Provisioning Framework of the SAP LT Replication Server system in order to consume data.

We recommend that the SAP LT Replication Server is installed on a separate system, but before you begin the installation, it is important to understand the various system landscape options available.

The SAP LT Replication Server can be installed either:

- On the source system
- On an SAP Solution Manager system or on a separate system
- On an SAP BW system.

The SAP LT Replication Server uses background processing to replicate data. This can be an important factor in deciding where to install SAP LT Replication Server since background processing uses CPU cycles. If the SAP LT Replication Server is a separate system, this ensures that the background processes do not run on the source system. This option separates the software maintenance activities (kernel upgrades/patch management and so on) from the source system.

Note

- The SAP LT Replication Server system must be a Unicode system
- Ensure that database performance is good. Poor database performance can result in poor SAP LT Replication Server performance.

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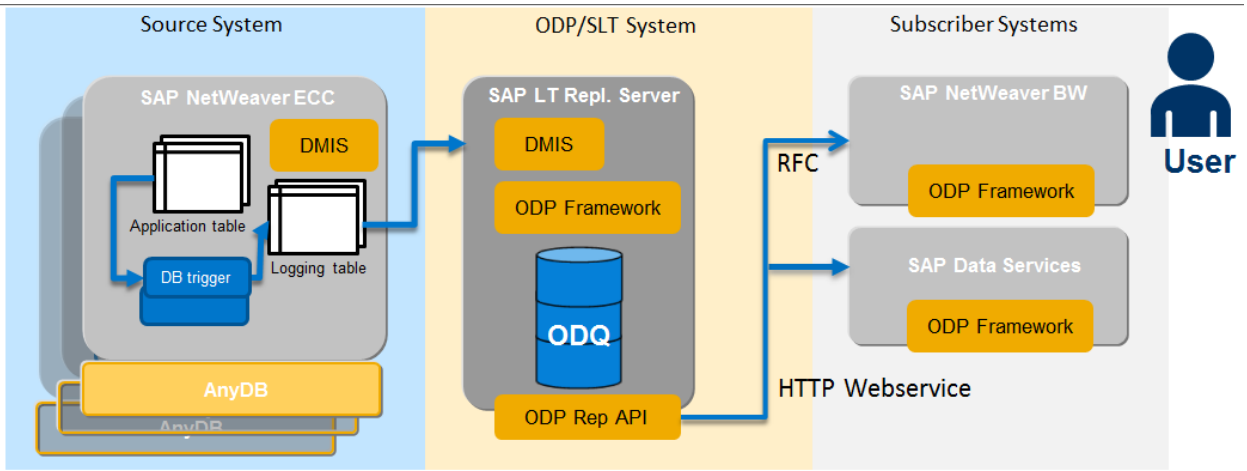
2.2 Installation of SAP LT Replication Server on a Separate System

We recommend that the SAP LT Replication Server is installed on a separate system as illustrated below:



Typical system landscape.

2.3 Comparison of Different Installation Options

The following table outlines in detail the advantages and disadvantages of the different installation options:

	Separate System	BW System	Source System (if SAP)
Advantages	• No software maintenance dependencies • Flexibility	• Simplified landscape and administration • Re-use of existing NW instance	• Simplified landscape and administration • Re-use of existing SAP ERP instance

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	Separate System	BW System	Source System (if SAP)
Disadvantages	• Investment and maintenance effort for separate server / NW instance	• Performance impact • Potential software maintenance dependencies	• Performance impact • Potential software maintenance dependencies

2.4 Compatibility Regarding SP Levels

2.4.1 DMIS Add-on

The table below outlines the compatibility between the different SP levels of the source system, the SAP LT Replication Server, and the subscriber application.

Source System	SAP LT Replication Server	Subscribers
DMIS 2011 SP10 or higher	DMIS 2011 SP13	SAP BW 7.3 – BW 7.50 (BW on HANA) SAP BO Data Services 4.2 SP1

Note

The DMIS add-on does not need to be installed in the system on which the subscriber application runs.

2.4.2 ODP Framework

The ODP framework is part of the SAP NetWeaver PI_BASIS component. To use the ODP scenario for real-time replication, the ODP framework is needed in the **ODP/SLT system** and the **subscriber systems**.

You can use any of the following PI_BASIS releases:

- 730 SP10 or SP8 (together with SAP Note 1848320)
- 731 SP9 or SP5 (together with SAP Note 1848320)
- 740 SP4 (or higher) or SP2 (together with SAP Note 1848320)

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The ODQ component which is needed in the ODP/SLT systems is part of SAP_BASIS component.

You can use any of the following SAP_BASIS releases:

- 730 SP10 or SP5 (together with SAP Note 1817467)
- 731 SP8 or SP3 (together with SAP Note 1817467)
- 740 SP4 (or higher) or RTC (together with SAP Note 1817467)

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3 Source System Preparation and Installation

3.1 Preparatory Steps for Source System

Overview

Use this section to check that the source system is suitable for the replication of data using SAP LT Replication Server. Note that the SAP LT Replication Server is shipped in a specific add-on (DMIS_2011* or DMIS_2010*).

If you intend to run SAP LT Replication Server on your source system, the details outlined in section 4.1, Preparatory Steps for SAP LT Replication Server System also apply. Note that you can use more than one source system to replicate data to SAP BW by using the ODP framework.

Overview of Required Components

- The DMIS Add-on, DMIS_2011 with a minimum support package 03 (we recommend using the latest available support package).

System Requirements

The following conditions apply for the source system:

- The system must be of at least SAP Basis release SAP NetWeaver 6.20.

To check whether your system is suitable for the installation of the DMIS add-on, DMIS_2011, proceed as follows:

1. Determine the SAP Basis version of your source system (choose System → Status, and choose the Other Kernel Info pushbutton)
2. SAP Note 1577441 and SAP Note 1577503 contain information about the installation and upgrade to DMIS 2011. Refer to these SAP Notes in order to determine that your SAP Basis and support package versions are supported.

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3.2 Installation Steps for Source System

Use this section to install the required DMIS component in your source system.

Note

If you plan to run the SAP LT Replication Server on the source system, you only have to follow the procedure that is described in chapter 4, SAP LT Replication Server Preparation and Installation.

All required software components are available from the SAP Support Portal https://support.sap.com/swdc and can be installed with the SAP Add-On Installation Tool (SAINT). For more information about SAINT, see here.

To install the DMIS Add-on (DMIS_2011), proceed as follows:

1. Navigate to the SAP Support Portal, https://support.sap.com/swdc
2. Download the relevant DMIS version from the following path:

   Installations and Upgrades → *A-Z Index*→ *L* → *SAP LT Replication Server* → *SAP LT Replication Server 2.0*
3. Follow the DMIS installation procedure as described in SAP Note 1577441. For information regarding the compatibility of the DMIS versions for the ODP scenario, see SAP Note 1914764.
4. Download latest available support packages from the following path:

   Support Packages and Patches → *A-Z Index* → *L* → *SAP LT Replication Server* → *SAP LT Replication Server 2.0*
5. Choose *Entry by Component*, and select the relevant product component version.

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4 ODP/SLT System Preparation and Installation

4.1 Preparatory Steps for SAP LT Replication Server System

Use this section to check that the SAP LT Replication Server system is suitable for the replication of data using SAP LT Replication Server. Note that the SAP LT Replication Server is shipped in a specific add-on (DMIS_2011* or DMIS_2010*).

> **Note**
>
> We recommend that the SAP LT Replication Server System is a separate system.

Overview of Required Components

- The DMIS add-on, DMIS_2011, with a minimum support package 10 (we recommend using the latest available support package)

- SAP NetWeaver 7.30 or higher with one of the following PI_BASIS components:
  - 730 SP10 or SP8 (together with SAP Note 1848320)
  - 731 SP9 or SP5 (together with SAP Note 1848320)
  - 740 SP4 (or higher) or SP2 (together with SAP Note 1848320)

System Requirements

The following requirements apply for the SAP LT Replication Server system:

- The system must be an SAP system with minimum SAP NetWeaver release of 7.30 (with at least Basis support package 8) ABAP stack

> **Note**

- File system: 100 GB
- RAM: 16-32 GB

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- CPU: 2-4 cores
- Minimum number of background jobs: 10

To check whether your system is suitable for the installation of the DMIS add-on, DMIS_2011, proceed as follows:

1. Determine the SAP Basis version of your SAP LT Replication Server system (choose System → Status, and choose the *Other Kernel Info* pushbutton)
2. SAP Note 1577441 and SAP Note 1577503 contain information about the installation and upgrade to DMIS 2011.

4.2 Installation Steps for SAP LT Replication Server System

Use this section to install the required DMIS component in your SAP LT Replication Server system.

All required software components are available from the SAP Software Download Center https://support.sap.com/swdc and can be installed with the SAP Add-On Installation Tool (SAINT). For more information about SAINT, see here.

To install the DMIS add-on, DMIS_2011, proceed as follows:

1. Navigate to the SAP Software Download Center, https://support.sap.com/swdc
2. Download the relevant DMIS version from the following path:

   Installations and Upgrades → *A-Z Index*→ *L* → *SAP LT Replication Server* → *SAP LT Replication Server 2.0*

   Follow the DMIS installation procedure as described in SAP Note 1577441.

3. Download latest available support packages from the following path:

   Support Packages and Patches → *A-Z Index* → *L* → *SAP LT Replication Server* → *SAP LT Replication Server 2.0*
4. Choose *Entry by Component*, and select the relevant product component version.

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5 Post Installation Activities

5.1 Activation of Web Dynpro and Relevant Services

After the installation of SAP LT Replication Server, all required Web Dynpro SAP LT Replication Server services are initially disabled. You must enable these services in order to run the SAP LT Replication Server user interface.

To activate the SAP LT Replication Server services, proceed as follows:

1. Activate Web Dynpro services as described here: Active Services in SICF
2. Run transaction SICF.
3. In the *Hierarchy Type* field, enter **SERVICE**, and choose *Run*.
4. Expand the node default_host, and navigate to sap → bc → webdynpro → sap
5. Activate the following services:
    - iuuc_replication_config
    - iuuc_repl_mon_powl
    - iuuc_helpcenter
    - iuuc_helpcenter_document
    - iuuc_repl_wdc_config_gaf
    - iuuc_repl_mon_schema_oif
    - /sap/public/bc
    - /sap/public/bc/ur
    - /sap/public/mysso/cntl
    - /sap/bc/webdynpro/sap/iuuc_repl_mon_schema_oif
    - /sap/public/bc/icons
    - /sap/public/bc/icons_rtl
    - /sap/public/bc/webicons
    - /sap/public/bc/pictograms
    - /sap/public/bc/webdynpro
    - /sap/public/bc/webdynpro/adobeChallenge
    - /sap/public/bc/webdynpro/mimes
    - /sap/public/bc/webdynpro/ssr
    - /sap/public/bc/webdynpro/ViewDesigner
    - /sap/bc/nwbc

We also recommend using SSL to protect the network communications. This requires the configuration of the underlying SAP NetWeaver ABAP Server for the HTTPS protocol, and that the security settings are set to 'SSL' for the above listed ICF services.

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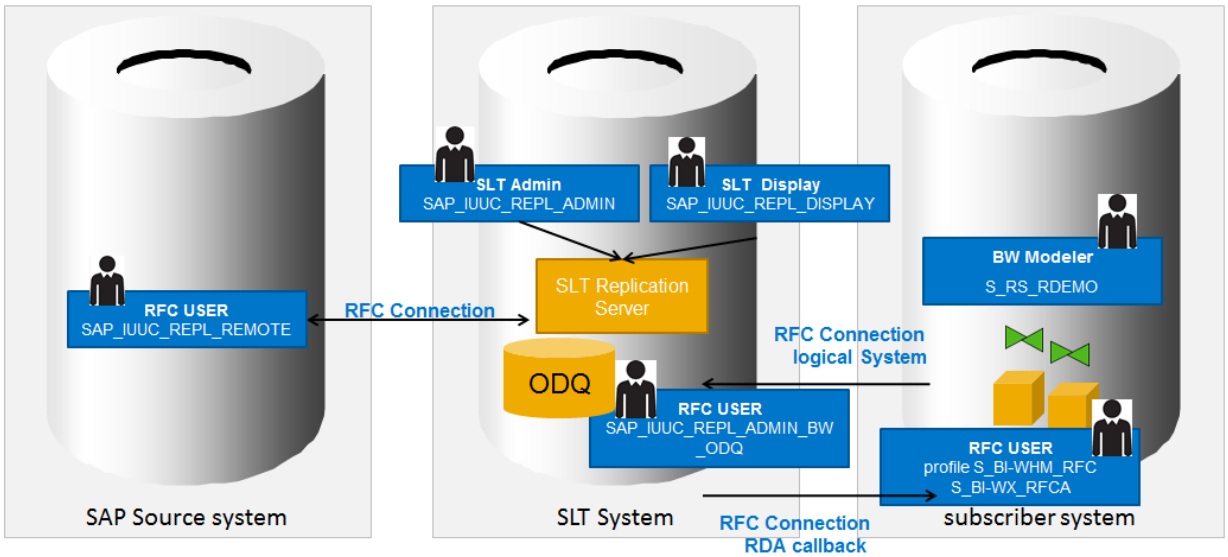
5.2 User Creation and Connections

The following figure illustrates the various users required in the source, subscriber, and SAP LT Replication Server systems. The following sections describe in detail exactly what is required.



5.2.1 User Creation and Connections for SAP Source System and ODP/SLT System

In order to replicate data using the SAP LT Replication Server, you must create an RFC connection to the source system and to the SLT system as the target system.

In order to do this, the following role is required:

- SAP_IUUC_REPL_REMOTE

Note

If you use a new client after the DMIS add-on is applied, you must transport the necessary roles from client 000 into your target client.

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Note

For more details about the roles and authorization concept of SAP LT Replication Server, see the Security Guide for SAP LT Replication Server.

Prerequisites

Refer to the SAP user administration guide for RFC user creation.

Generating Role SAP_IUUC_REPL_REMOTE

To generate role SAP_IUUC_REPL_REMOTE, proceed as follows:

1. Execute transaction **PFCG**.
2. In the role field, enter the role **SAP_IUUC_REPL_REMOTE**.
3. Choose the *Change Role* pushbutton.
4. On the *Authorizations* tab page, choose the *Change Authorization Data* pushbutton.
5. The system displays the *Change Role: Authorizations* screen. Choose the *Generate* pushbutton.
6. Return to the *Authorizations* tab page; this tab page should now have a green light.
7. In the *User* tab page, choose the *User Comparison* pushbutton.
8. The system displays the *Compare Role User Master Record* screen. Choose the *Complete Comparison* pushbutton.
9. The *User* tab page should now also have a green light.

Creating Users and RFC Connections

Create the required user and RFC connection as follows:

1. Create a user (of type *Dialog* or *System*) in your source system and in the SAP LT Replication Server (target) system, generate and assign the following role to this user:
   - SAP_IUUC_REPL_REMOTE

     Note

     Do not use user DDIC. The role SAP_IUUC_REPL_REMOTE is not generated by default. You must generate and assign this role to the newly created user.
2. Create an RFC connection (type 3 – ABAP) from the SAP LT Replication Server to the source system with the newly created user (if both systems are Unicode, specify the RFC connection as Unicode).

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3. Create an RFC connection from the SAP LT Replication Server system that points to itself (the target system).
4. In the SAP LT Replication Server system, create a user with the role SAP_IUUC_REPL_ADMIN_BW_ODQ. This user is needed later for the connection from the BW system to the ODQ (generate this role as explained for SAP_IUUC_REPL_REMOTE).

5.2.2 User Creation for SAP LT Replication Server System

The SAP LT Replication Server is delivered with an own role SAP_IUUC_REPL_ADMIN. To activate the role, follow the procedure described in 6.2

For more details about the roles and authorization concept of SAP LT Replication Server, see the Security Guide for SAP LT Replication Server.

5.3 Separate Tablespace for Logging Tables

It is possible (but not essential) to store the source system replication log tables in a separate table space. The decision to do this is the responsibility of the system administrator. One advantage of having the log tables in their own table space is that you can easily monitor the size of the log tables.

As each database system has its own method of providing this functionality, refer to your database documentation for this procedure.

If you use own data classes and tablespaces, see SAP Note 46272.

Chapter 6.5.1 describes how to make your tablespace known to the configuration in the SAP LT Replication Server.

5.4 Creating a Configuration

In the SAP LT Replication Server system, you define a connection between the source system and the SAP LT Replication Server as the target system. You specify this information as a configuration.

To create a configuration, proceed as follows:

Run transaction **LTRC**. Choose the *New* pushbutton. Creating a configuration involves the following steps:

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Specify General Data

In this step, you specify the following information:

- Configuration Name
  - You must specify a name for the configuration. This name is used for Operational Delta Queue.
- Description (Optional)
  - You can specify a description for the configuration.
- Authorization Group (Optional)
  - If you have sufficient authorizations, and require specific authorizations for the configuration, you can specify an authorization group here, and use this authorization group in the corresponding authorization object S_DMIS_SLT.

Specify Source System

For SAP source systems, you specify the following information:

- System Data
  - Specify an SAP system as a source system.
- RFC Destination
  - Specify the RFC destination to the source system.
- Allow Multiple Usage
  - Select this checkbox if you want to replicate data from an SAP source system to multiple target systems.
- Read from Single Client
  - Select this checkbox if you only want to replicate data from the client that is specified in the RFC destination.

Specify Target System

Enter your SLT system as target system and specify the following information:

- RFC Destination
  - Specify the RFC destination to the target system (or enter NONE).
- Scenario for RFC Communication

  Choose the following replication scenario:
  - **ODP Replication Scenario**

    When you select the ODP Replication Scenario, the system automatically checks if the BAdI implementation for ODP (enhancement implementation ODQ_ENH_SLT_REPLICATION) is active or not. If the BAdI implementation is not yet active, you must activate it after you have created the configuration.

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Enter a 6-digit queue alias that will be the identifier for source system tables for this configuration. In SAP BW, you must specify an ODP context. This ODP context corresponds to the queue alias. The naming convention in SAP BW is SLT~<6 digit queue alias>.

Specify Transfer Settings

In this step, you specify the following information:

- Initial Load Mode
  - There are different options (reading types) available for the initial load. These reading types access the data in the source system table in different ways. In order to accelerate the initial load, you can change the data load behavior for the entire configuration. The default setting is *Resource Optimized* (reading type 3 for all tables), but you can accelerate the data load by choosing *Performance Optimized* (reading type 5 for transparent tables, and reading type 4 for cluster tables).

Information about Reading Types:

- Reading Type 3 - DB_SETGET

The reading type DB_SETGET is the default reading type. It uses a function to read the records of the source table ordered by the primary key. This means that no additional index is required as the primary index is used. The function returns a set of table records (the default is 10,000 records).

- Reading Type 4 and 5 - Index Cluster

Data is read from the source table and stored in a cluster-like table (DMC_INDXCL) in the source system. All rows are read from the source table, divided into portions and compressed in a raw format. Because data is extracted to a separate table in the source system, additional temporary tablespace is required for table DMC_INDXCL (approximately 10% of the source table size for transparent tables and between 50% to 75% of the source table size for cluster tables). Reading type 5 is the same as reading type 4, but uses a full table scan to read the data, (the preferred method for reading data from transparent tables).

- Dataclass of Tablespace
  - You can specify a tablespace for the logging tables in the source system.
- No. of Data Transfer Jobs
  - You can specify the number of jobs that are used for the data transfer process in the SAP LT Replication Server. This value specifies the number of data transfer jobs which will run in the SAP LT Replication Server to replicate the tables of the RFC connection to the target system. For more information, see the Application Operations Guide (note that this guide details specifics for SAP LT Replication Server for SAP HANA, but the principles described here with regard to data transfer jobs are valid for the BW context).
- No. of Initial Load Jobs
  - You can specify the number of jobs that are used for the initial load in the SAP LT Replication Server.
- No. of Calculation Jobs

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- For reading types 1 and 3, this is the number of jobs that are used to calculate the data transfer portions that are used for the initial load. For reading types 4 and 5, it is the number of jobs that transfer the portions to table DMC_INDXCL.

- Replication Options

You can choose one of the following replication options:

- Real-time - The trigger-based data replication method that continuously replicates any data changes to the target system.
- Schedule by Interval - You can specify a time and frequency for the replication. For example, every 30 minutes or every 12 hours.
- Schedule by Time - You can specify a specific time for the replication, for example 23:00. The SAP LT Replication Server would then replicate any database changes to the target system every day at 23:00.

Application (Optional)

Depending on your use case, you may be required to specify an application.

Review and Create

In this step, you can review your settings and create the configuration.

Note

In order to run the ODP replication scenario for SAP LT Replication Server, a specific BAdI implementation for ODP (enhancement implementation ODQ_ENH_SLT_REPLICATION) is required. The BAdI must be active in order to run the replication scenario.

To check whether the BAdI is active, proceed as follows:

1. Open transaction LTRC, and choose the configuration that you have created.

2. Open the tab page *Expert Functions* and select *ODP/ODQ Specific Functions* → *Check whether BAdI Implementation is Active*.

3. If the system response shows that the BAdI is inactive, navigate back to the expert functions and select *Activate/Deactivate BAdI Implementation*.

5.5 Data Replication Triggered from Subscriber System

The replication of data from the Operational Delta Queue is triggered from subscribing SAP applications. The subscribing application uses an RFC connection to request the data from the SAP LT Replication Server system/ODP system.

An RFC user is required in the SAP LT Replication Server system with the role AP_IUUC_REPL_ADMIN_BW_ODQ.

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> **Note**
>
> For information about role generation refer to chapter 5.2, User Creation and Connections.

The process for triggering data replication from the subscriber system is as follows:

1. The subscriber has to create a connection to the ODP system with the predefined ODP context. During this setup, the following two connections are established:
   - The RFC connection from the BW to the SAP LT Replication Server system (you have to create this connection)
   - The RFC connection <BW system ID>CLNT<client of BW system> is created automatically in the SLT system with the standard BW background user.

   > **Note**
   >
   > Check that the correct BW Background user is used (standard: BWREMOTE), and that you enter the correct password.

   > **Note**
   >
   > This step is described for a subscribing SAP NW BW system in the help portal documentation:
   >
   > **SAP NetWeaver 7.3 - Including Enhancement Package 1** → Application Help → Function-Oriented View → <Your Language> → SAP NetWeaver Library: Function-Oriented View → Business Warehouse → Data Warehousing → Modeling → Data Acquisition → Transferring Data Using Operational Data Provisioning → Creating an ODP Source System.

2. The subscriber creates DataSources for the ODP source system and an InfoPackage or DTP to start the data transfer.

   For more information about the data transfer setup, see the following demo:
   http://demo.tdc.sap.com/SpeedDemo/63a6390d735b90b2

After you have created and activated the DataSource, you can preview the data to check if the fields and contents are transferred correctly (pushbutton *Read Preview Data* in the tab page *Preview*)

Hint: For a quick preview, enter a small number of records that you want to display instead of transferring the full table content.

5.6 Advanced Replication Settings

You can use transaction LTRS to specify advanced replication settings. For example, you can:

- Modify target table structures
- Specify performance optimization settings

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- Define transformation rules to transform data from the source table during the replication process. The transformed data will then be present in the target table. For example, you can specify rules to convert fields, fill empty fields, and skip records.
- Assign views to tables (database views and projection views). For more information, see the SAP Community Network under http://scn.sap.com/community/replication-server.

For more information about these advanced replication settings, see the documentation that is accessible from the UI of transaction LTRS.

5.7 Accessing the Configuration and Monitoring Dashboard

You use the *Configuration and Monitoring Dashboard* to view status information for the replication.

You can access the *Configuration and Monitoring Dashboard* by using transaction **LTR**.

Note

In order to ensure compliance with security standards, the Configuration and Monitoring Dashboard (transaction **LTR**) requires the use of the SAP NetWeaver Business Client 3.5 or 4.0.

The reason for this is that the SAP NetWeaver Business Client supports a logout for all Web Dynpro windows. Web browsers do not support this logout feature. For example if you use a web browser to access the work center, there is no option to logout. Simply closing the web browser window does not log the user out of the system. The session runs on the server until it times out, and this is a potential security risk.

In order to use the SAP NetWeaver Business Client, the relevant user must have the role SAP_IUUC_REPL_NWBC assigned to them.

Note that you can also use the SAP LT Replication Server Cockpit (transaction LTRC) in order to work with configurations.

5.8 Monitoring of Data Transfer and the Operational Delta Queue

Monitoring of this scenario has several aspects - the functionality of the data transfer infrastructure, the data in the Operational Delta Queue, and the correct transfer of the data into the DataSource of the subscriber:

- You can use the **SAP LT Replication Server Cockpit** (transaction *LTRC*) to control the replication process for the SAP LT Replication Server and to use the monitoring functionality like for the running jobs (but you cannot perform actions in the cockpit like start/stop/suspend/resume the data provisioning) .

  For more information on the SAP LT Replication Server Cockpit (transaction *LTRC*), refer to the help documentation that is available from the UI.

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- You can monitor the Operational Delta Queues in the ODP/SLT system by using transaction **ODQMON**. You can view the queues, the corresponding requests, and the amount of contained data. If there are extraction errors, you can view them in transaction ODQMON. To do this, select a subscription in the *Monitor Delta Queue Subscriptions* screen and choose the *Display Extraction Errors for Delta Queue* pushbutton.

For more information on ODQMON, refer to the SAP Help Portal documentation on http://help.sap.com/ → Technology Platform → SAP NetWeaver → SAP NetWeaver Platform → SAP NetWeaver 7.4 → Application Help → Function-Oriented View → <Your Language> → Search and Operational Analytics → Operational Data Provisioning → Data Extraction and Data Replication with Operational Data Provisioning → Monitoring Delta Queues.

- If the subscriber is a SAP BW system, you can view the replicated data in the SAP BW system by using the **Administrator Workbench** (transaction RSA1).

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6 Resources for Operational Information

6.1 SAP Data Services

For information about the new features available with SAP Data Services release 4.2 Support Package 1, see here.

For the Supplement for SAP guide, see here.

In particular, see chapter 4, Connecting to SAP Applications, and section 4.2, Reading from SAP ODP Sources.

6.2 SAP BW

For information about transferring Data with SAP Landscape Transformation Replication Server, see here.

For information about transferring data from SAP Landscape Transformation Replication Server using Operational Data Provisioning, see here

6.3 Operational Data Provisioning

For information about monitoring delta queues, see here.

For information about monitoring requests in the Delta Queue Monitor (including hints for troubleshooting), see here.

For information about setting the retention period for data in the delta queue, see the documentation in transaction **ODQMON** under *Goto* → *Reorganize Delta Queues*.

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6.3.1 Deleting a Subscription

In the SAP Landscape Transformation Replication Server system, you can use transaction ODQMON (Monitor Delta Queue) to delete a subscription. In the *Queue* field, you can specify the subscriptions that are relevant for your configuration. In the Request Selection field, ensure that you specify the value All, so that all queues are selected. To view the subscriptions, choose the *Subscriptions* pushbutton. To delete a subscription, select the relevant subscription and choose the *End Subscription* pushbutton.

If you want to end a subscription in the subscriber (BW) system, you have to delete all delta requests of the DTP (Data Transfer Process) of the InfoProvider and then delete the DTP itself. For more information about request handling, see Loading Data from ODP Source Systems Directly into InfoProviders.

6.3.2 Additional Information about Subscriptions

For specific information about a subscription, you can refer to the following tables:

- IUUC_ODQ_ALIAS

  You can use this table to view the mass transfer ID and the ODQ alias. The ODQ alias corresponds to the ODP context.
- IUUC_ODQ_TABLES

  This table contains all incoming requests. All requests are assigned a new request ID (a unique timestamp). You can view the relevant table and mass transfer ID for the request. You can also view the extraction mode for the request, and whether a request is active or not
- IUUC_ODQ_PARAM

  This table contains information about any constraints, if only a subset of the data was selected, for example by specifying a calendar year, you can view that information here.
- IUUC_ODQ_PRJ

  If only specific columns are part of the transfer, then you can use this table to view this information.

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7 References

7.1 List of Documents

The following table lists all documents mentioned in this Master Guide.

Title	Where to Find
Security Guide – Replicating Data to SAP HANA	http://help.sap.com/sapslt
Application Operations Guide – Replicating Data to SAP HANA	http://help.sap.com/sapslt (Specifically, chapter 3.5, Sizing and Basic Job Handling)

7.2 List of SAP Notes

The following table lists all SAP Notes mentioned in this Master Guide.

SAP Note Number	Title	Description
2446237	Installation/Upgrade SLT - DMIS 2011 SP12	This SAP Note describes the installation or upgrade of SAP LT Replication Server to the relevant DMIS SP.
19466	Downloading SAP Kernel patches	Downloading a kernel patch in the Software Distribution Center.
517484	Inactive services in the Internet Communication Framework	The Internet Communication Framework Services are inactive when you install the SAP Web Application Server.
1577441	Installation and delta upgrade of DMIS 2011_1	This SAP Note contains information about installing DMIS_2011 and upgrading to DMIS_2011

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SAP Note Number	Title	Description
1577503	Upgrade to DMIS 2011_1 in the System Switch Upgrade	This SAP Note contains information about upgrading to DMIS_2011
2037476	Operational Data Provisioning with SAP LT Replication Server	This SAP Note describes the ODP scenario with SAP LT Replication Server and provides installation details.

7.3 SAP Help Portal Documentation

For more information about transferring data with SAP LT Replication Server, see SAP Library for SAP NetWeaver BW on SAP Help Portal at http://help.sap.com under → SAP NetWeaver → SAP NetWeaver Platform → <Choose the Relevant Release>:

- **SAP NetWeaver 7.4** → Application Help → Function-Oriented View → <Your Language> → SAP Business Warehouse → Data Warehousing → Modeling → Data Acquisition Layer → Data Provision Using Source Systems → Transferring Data with SAP Landscape Transformation Replication Server → Transferring Data from SLT Using Operational Data Provisioning
- **SAP NetWeaver 7.3 - Including Enhancement Package 1** → Application Help → Function-Oriented View → <Your Language> → SAP NetWeaver Library: Function-Oriented View → Business Warehouse → Data Warehousing → Modeling → Data Acquisition -> Transferring Data with SAP Landscape Transformation Replication Server
- **SAP NetWeaver 7.3** → Application Help → Function-Oriented View → <Your Language> → SAP NetWeaver Library: Function-Oriented View → Business Warehouse → Data Warehousing → Modeling → Data Acquisition → Transferring Data with SAP Landscape Transformation Replication Server

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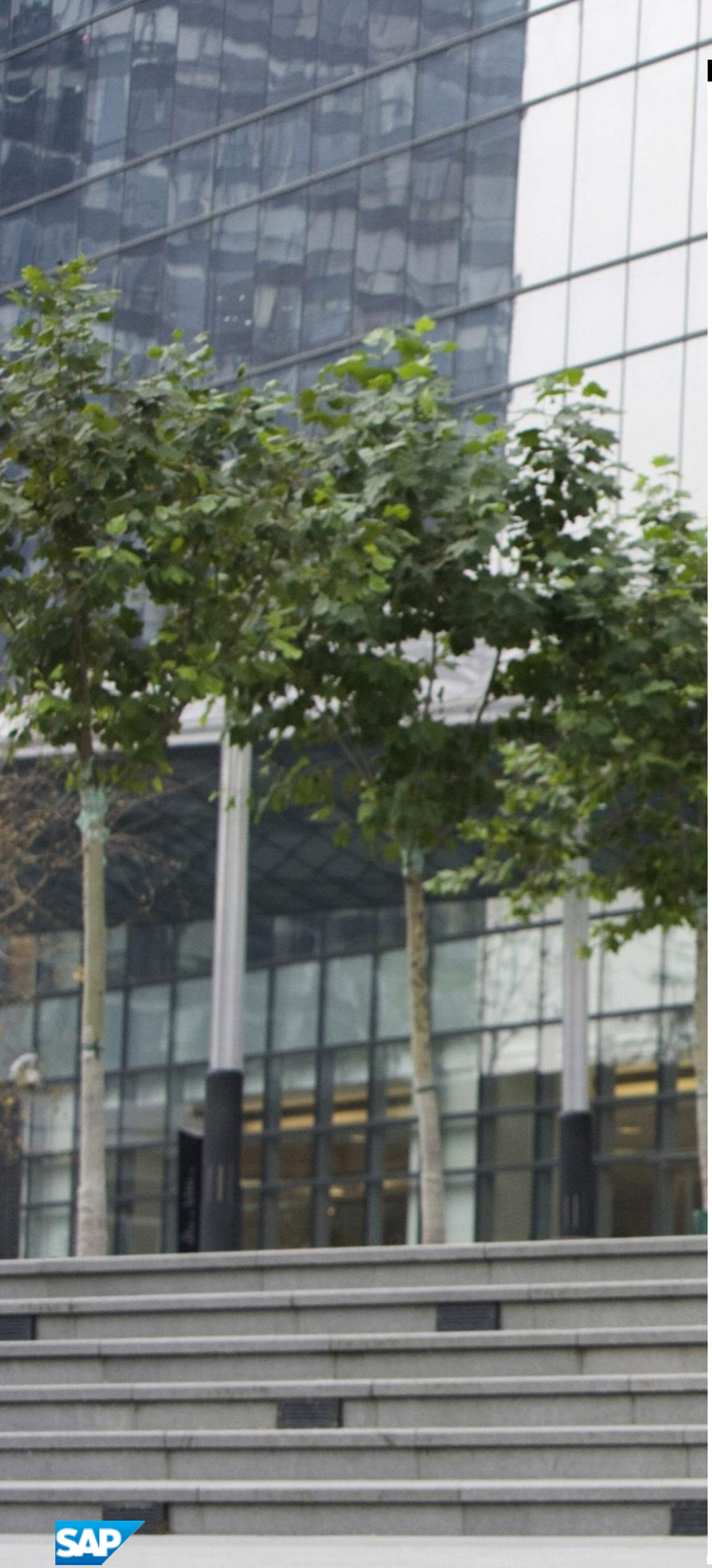

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